

Assignment 4: NLP Semantic Search Pipeline

1. Objective

The objective of this assignment is to build a Natural Language Processing (NLP) pipeline that processes raw text data and performs semantic similarity search using vector representations.

2. Dataset

The dataset used is the IMDb movie review dataset. It contains textual reviews which were used for preprocessing and similarity analysis.

3. Methodology

3.1 Text Cleaning

All text data was cleaned by:

- Converting to lowercase
- Removing punctuation and special characters

3.2 Tokenization

The cleaned sentences were split into individual words using tokenization.

3.3 Stopword Removal

Common words such as “the”, “is”, and “and” were removed to focus on meaningful words.

3.4 Lemmatization

Words were converted to their base/root form using lemmatization.

3.5 TF-IDF Vectorization

The processed text was converted into numerical vectors using TF-IDF, which captures word importance.

3.6 Word Embeddings

Word2Vec was used to generate vector representations of words based on context.

3.7 Sentence Embeddings

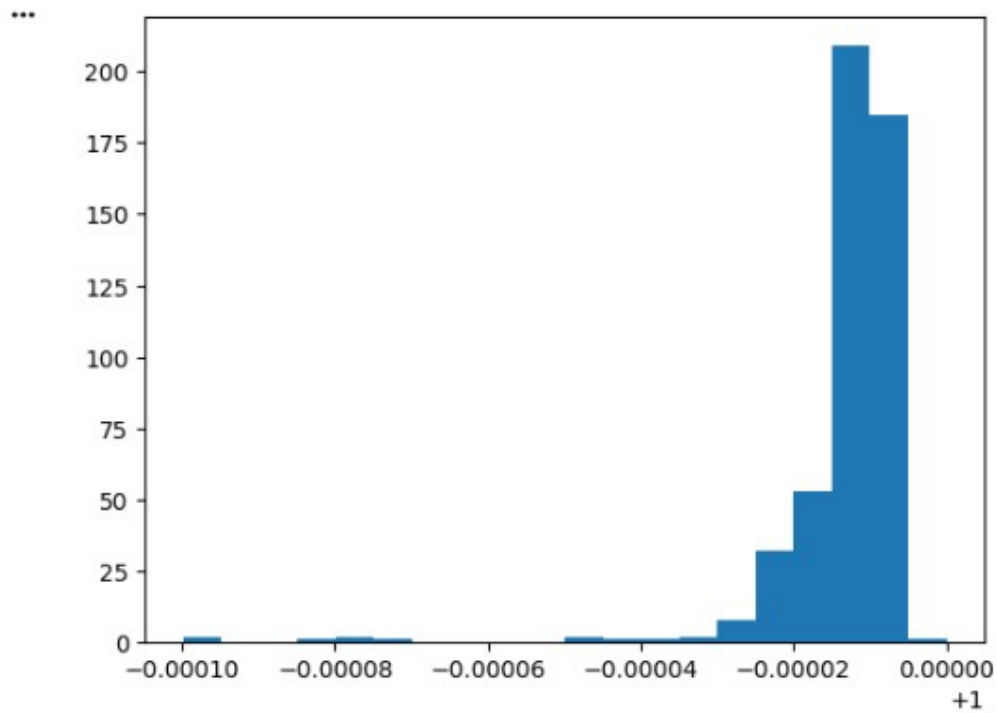
Sentence vectors were created by averaging word vectors to represent full sentences.

3.8 Similarity Calculation

Cosine similarity was used to compare sentence vectors and find similar sentences.

3.9 Visualization

A histogram was plotted to visualize similarity score distribution.



4. Results

The model successfully identified similar sentences based on semantic meaning.

Example:

Query Sentence:

"I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that surrounded it..."

Most Similar Sentence:

"It figures this is a French film, LOL, with the emphasis on young girls..."

The model was able to capture contextual similarity between sentences.

Query: I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that surrounded it w

Most similar: It figures this is a French film, LOL, with the emphasis on young girls with much older men...why is

Score: 0.9999949

5. Conclusion

The NLP pipeline effectively transformed raw text into meaningful vector representations and enabled semantic search. This approach can be applied in recommendation systems, search engines, and text analysis tasks.

6. Tools & Libraries Used

- Python
- NLTK
- Scikit-learn
- Gensim
- HuggingFace Datasets
- Matplotlib